

NeuroSurv

A new surveillance system for neurological infectious diseases in Belgium

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Introduction

Lyme neuroborreliosis (LNB) reporting became mandatory in Europe in 2019 (ECDC case definition). However, existing surveillance systems did not allow to report ECDC-compliant cases in Belgium.

Therefore, a new system, NeuroSurv, was launched in January 2024. In addition to reporting of LNB, it is used to collect data on tick-borne encephalitis (TBE) and acute flaccid paralysis (AFP).

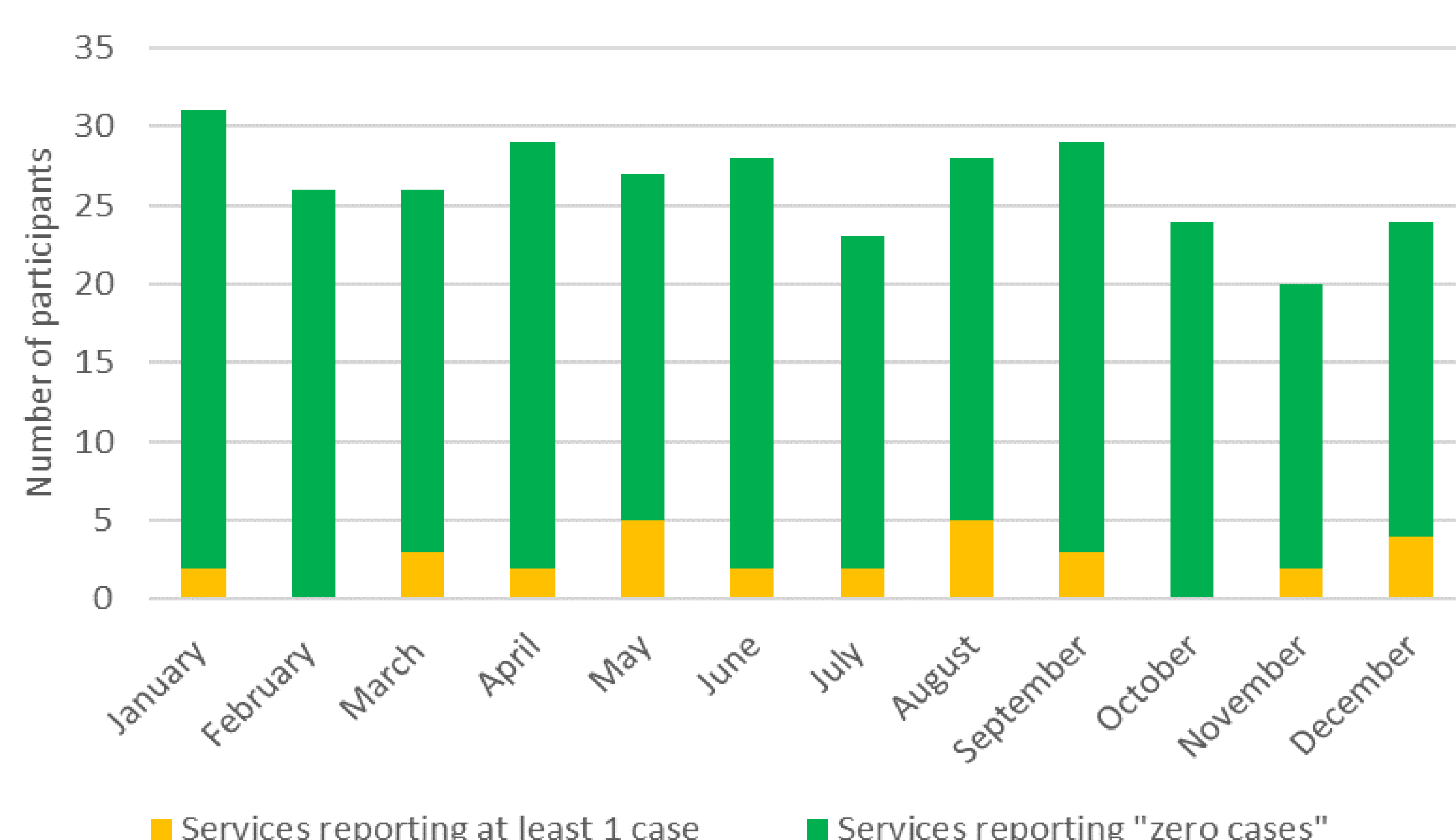
Methods

- **Voluntary participation** of (pediatric) neurology departments
- Monthly reporting via standardized LimeSurvey questionnaire
- **Aggregated data collected on LNB, TBE and AFP:**
 - Case numbers
 - Demographics, diagnostic tests, treatments, clinical signs
 - Zero-reporting is requested
- **Only ECDC-compliant LNB cases are reported to ECDC**

Results

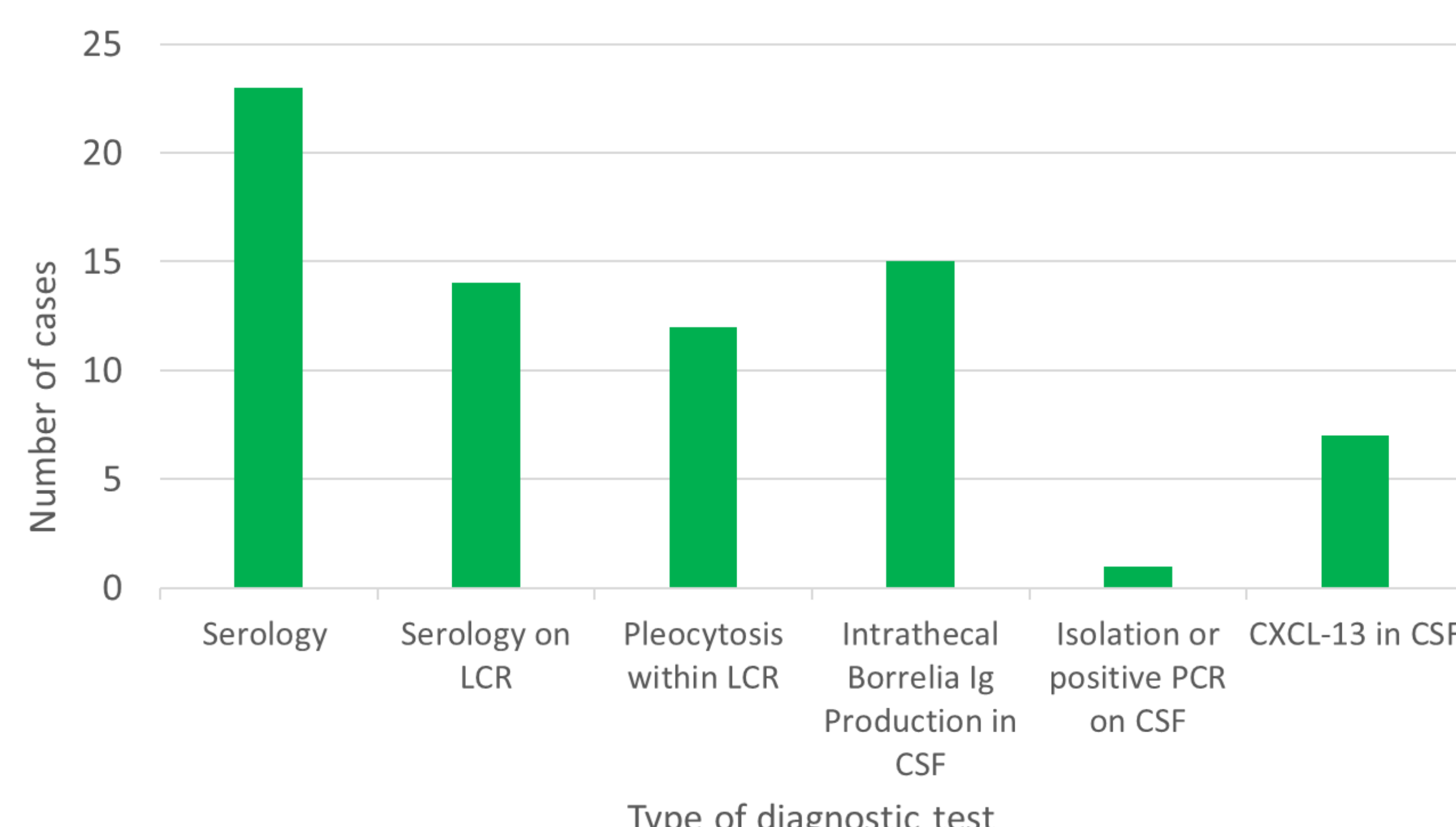
- **Participation:** 45 out of 121 hospitals contacted (**37%**)
- **Reporting:** **55%** average monthly participation, among which **90%** reported “zero cases”
- In 2024, **25 LNB cases** have been reported, among which **17 cases** (68%) met **ECDC criteria** for a probable or confirmed case

Fig. 1. Number of participating services to NeuroSurv, by month, 2024



- **Lumbar puncture** was performed for 19 out of 25 LNB cases (76%)
- **Age:** all LNB patients were **above 45 years old**
- **TBE and AFP cases:** 1 imported TBE case and 10 AFP cases were also reported in 2024

Fig. 2. LNB diagnostic tests, NeuroSurv, 2024 (multiple tests possible for one patient)



Challenges and perspectives

- In 2024, NeuroSurv allowed to report 17 LNB cases to ECDC. This is much lower than the annual estimated 270 LNB cases in Belgium^a.
- Several factors may contribute to this: not all LNB cases are diagnosed by a neurologist; non adherence to the recommendations for LNB diagnostic tests; voluntary participation; low and/or irregular participation rate.
- However, NeuroSurv successfully established a network of previously uninvolved physicians in surveillance, and raised awareness about neurological infectious diseases.
- Future efforts will focus on improving participation, promotion of the adherence to the guidelines for diagnosis of LNB, and collection of case-based data.

REFERENCES

a. Geebelen L et al. Combining primary care surveillance and a meta-analysis to estimate the incidence of the clinical manifestations of Lyme borreliosis in Belgium, 2015-2017. Ticks Tick Borne Dis. 2019 Apr;10(3):598-605. doi: 10.1016/j.ttbdis.2018.12.007

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